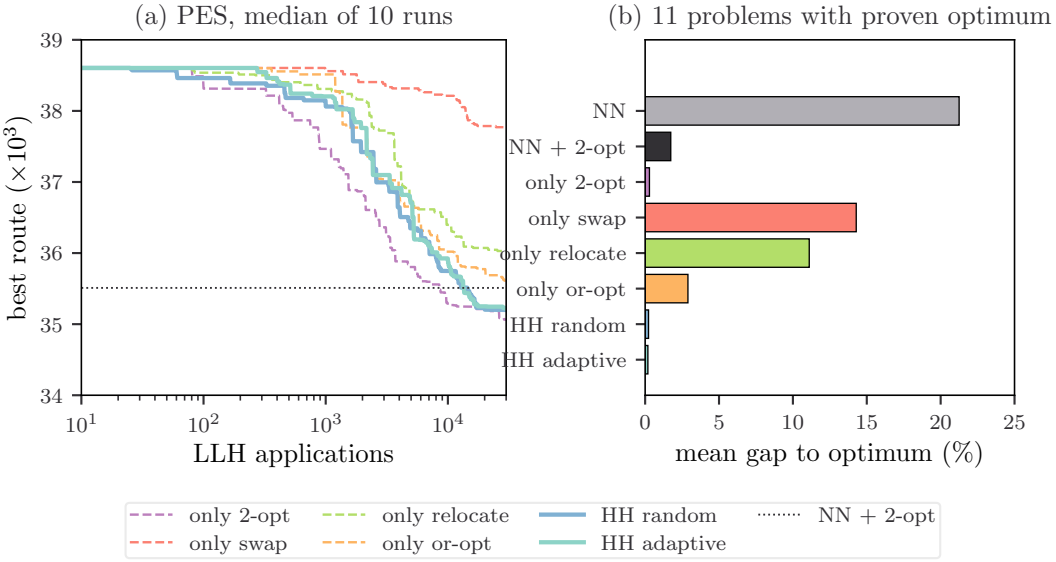

(a) PES, median of 10 runs
(b) 11 problems with proven optimum
best route (×10^3)
39
38
37
36
35
34
10^1
10^2
10^3
10^4
LLH applications
NN
NN + 2-opt
only 2-opt
only swap
only relocate
only or-opt
HH random
HH adaptive
0
5
10
15
20
25
mean gap to optimum (%)
only 2-opt
only swap
only relocate
only or-opt
HH random
HH adaptive
NN + 2-opt